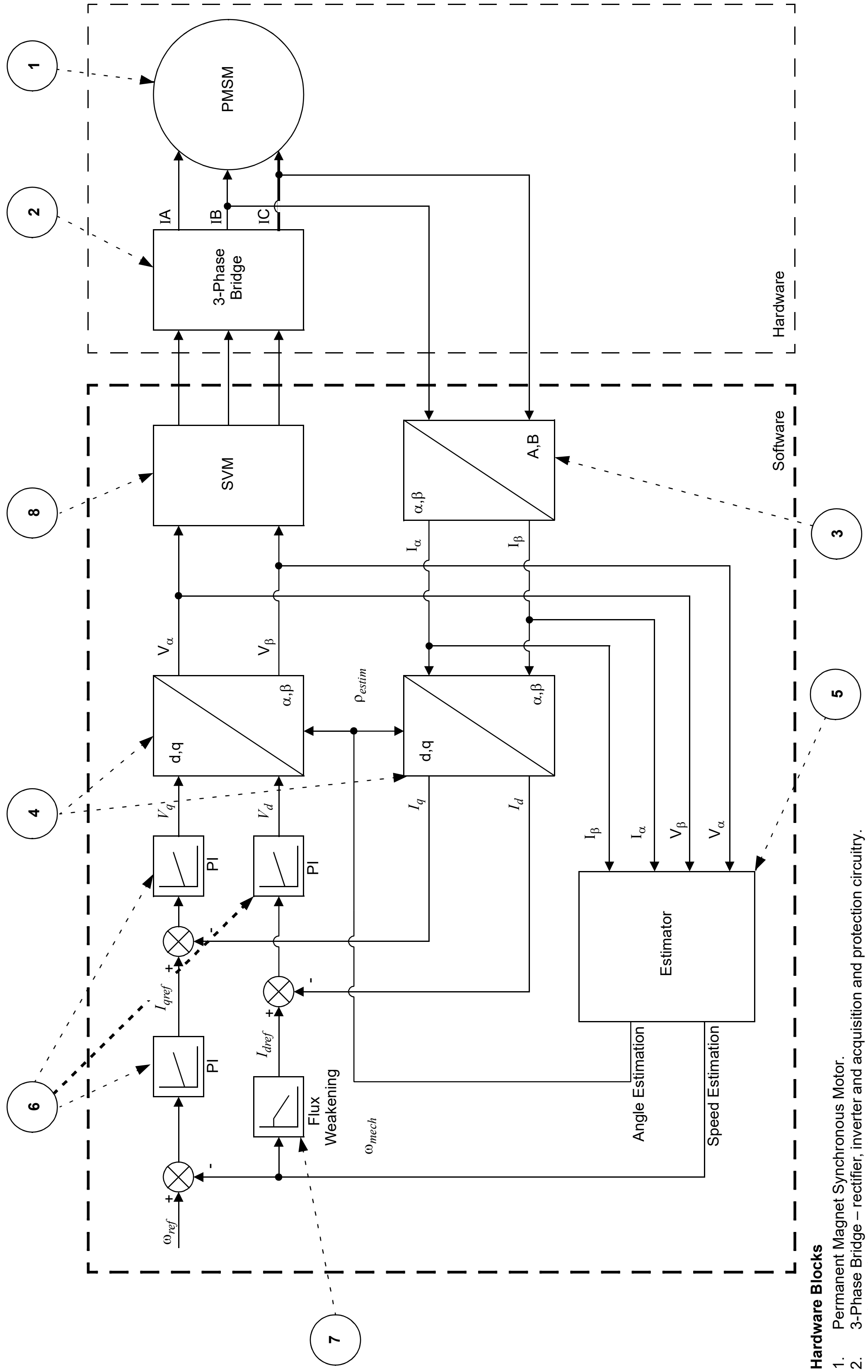
Figure 1-1. Sensorless FOC for PMSM Block Diagram



Hardware Blocks

1. Permanent Magnet Synchronous Motor.
2. 3-Phase Bridge – rectifier, inverter and acquisition and protection circuitry.

software (run by the device).

3. Clarke forward transform.
4. Park forward and inverse transform.
5. Angle and speed estimator.
6. Proportional integral controller.
7. Flux weakening.
8. Space vector modulation.